

# Unit 12, Lesson 7: Solving One-Step Equations in One Variable Using Addition and Subtraction – Part 1

## Equation

### Benchmark

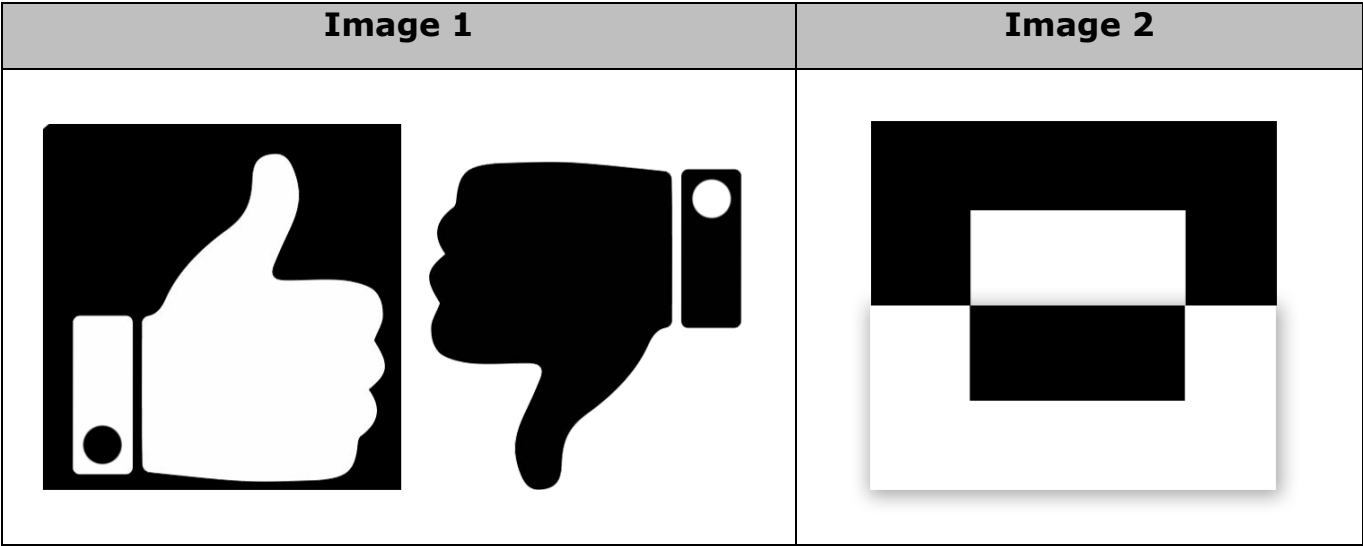
MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers.

### Learning Target

- ☐ I can solve addition and subtraction one-step equations in one variable using manipulatives, drawings, and number lines to show the inverse operations.

12.7.1 Warm-Up: Are They Inverses?

Angelica and Felipe were working on a project in art class. They were given the images below and instructed to explain whether or not the images are inverses.



Angelica said, “This was just like inverse operations from math class. Inverse means to do the reverse.”

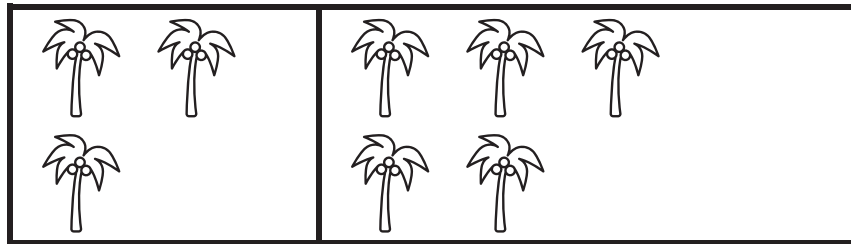
Felipe responded, “But the second image is a smaller image inside of the same image, so it is not a picture of inverses.”

1. Explain whether you agree with Angelica, Felipe, or both.
  
  
  
  
  
  
  
  
  
  
2. How do you see inverses represented in either image?

## 12.7.2 Guided Instruction: Solving Equations With Inverse Operations

Bar models and balance scales are useful tools for representing equations.

1. In the bar model shown, each palm tree represents 1 unit.

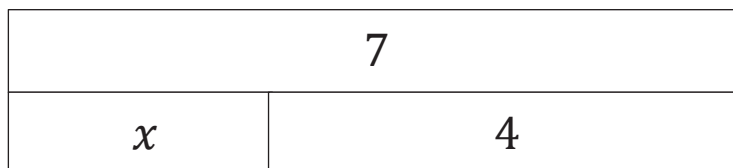


The bar model can be used to represent either an addition or subtraction equation. Fill in the missing values in each equation to match the bar model.

$$3 + \underline{\quad} = 8$$

$$8 - 3 = \underline{\quad}$$

2. The bar model shown can be used to represent either an addition or subtraction equation.



- a. Circle the addition equation that matches the diagram.

$x + 7 = 4$

$7 = x + 4$

$x = 7 + 4$

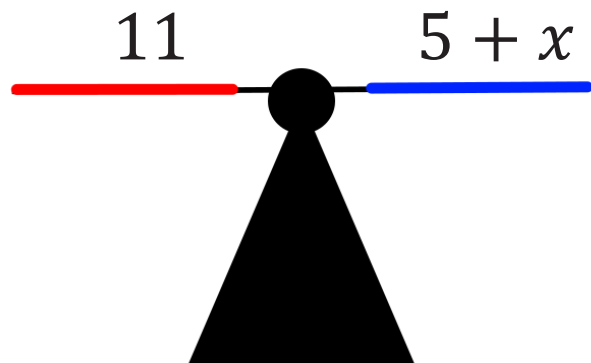
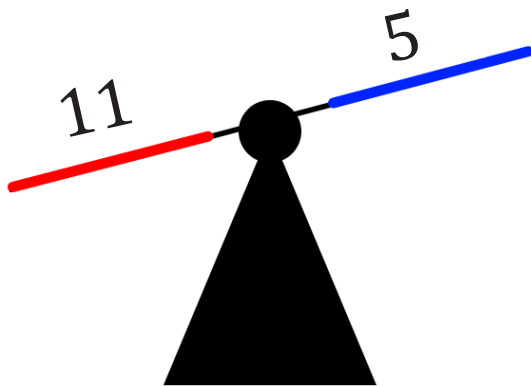
- b. Write an equivalent equation using the inverse operation.

c. Justify to your partner that the equation you wrote is an equivalent equation.

d. What is the value of the variable,  $x$ ?

3. Two balance scales are shown. The scale on the left is unbalanced. To balance the scale, an unknown value,  $x$ , needs to be added to the right side of the scale.

4.



a. The equation represented by the balance scale on the right is shown. Use inverse operations to find the value of  $x$  by completing the steps shown.

$$x + 5 = 11$$

$$x + 5 - \boxed{\phantom{00}} = 11 - \boxed{\phantom{00}}$$

$$x =$$

- b. Rewrite the original equation, substituting the solution for  $x$ .
- c. Explain how to use the equation in Part B to confirm that your solution is correct.

Inverse operations can be used to find the value of the variable in an algebraic equation by creating equivalent equations.



**Addition Property of Equality**

If  $a = b$ , then  $a + c = b + c$

**Subtraction Property of Equality**

If  $a = b$ , then  $a - c = b - c$

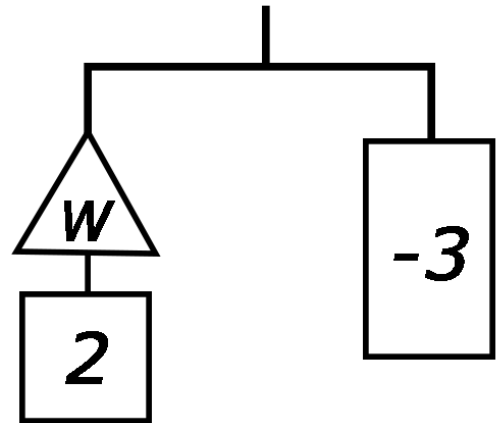
5. The equation  $x + 23 = 78$  can be solved by using inverse operations.
- a. Fill in values in each equivalent equation to solve for  $x$ .
- $$x + 23 = 78$$
- $$x + 23 - \underline{\hspace{2cm}} = 78 - \underline{\hspace{2cm}}$$
- $$x = \underline{\hspace{2cm}}$$
- b. What is the inverse operation used to solve the equation  $x + 23 = 78$ ?
- c. With your partner, discuss how to confirm whether or not the solution is accurate.

### 12.7.3 Guided Practice: Solving Equations With Inverse Operations

Work with your partner to complete the following.

An equation can also be represented using a balanced hanger. A hanger stays balanced when the weights on both sides are equal. As the weights are changed, the hanger will stay balanced as long as both sides are changed in the same way.

1. The hanger diagram represents a balanced equation because both sides of the hanger are level.



- a. Fill in the values for each equation using the hanger diagram. Then, solve for the unknown value,  $w$ .

$$\begin{array}{rcl}
 \triangle & + & \square = \square \\
 \triangle & + & \square - \square = \square - \square \\
 \triangle & = & 
 \end{array}$$

- b. Rewrite the original equation replacing the variable,  $w$ , with the solution to check your work.

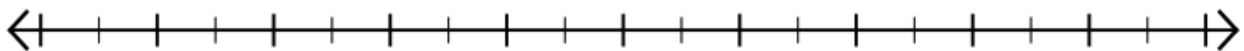
2. The bar model represents an equation.

|      |      |
|------|------|
| $x$  | $-4$ |
| $-9$ |      |

- Write an addition equation represented by the bar model.
- When solving an algebraic equation, the inverse operation can be applied to both sides of the equation. What inverse operation can be used to solve this equation?
- Write the equivalent equation using inverse operations.
- What is the value of the variable,  $x$ ?

3. Number line models can be used to represent and solve equations.

- Use the number line to represent the solution  $-6 = -5 + n$ .



- What is the value of the variable,  $n$ ?

**12.7.4 Your Turn!**

1. The equation  $-31 = s - 6$  can be solved using inverse operations.

a. Fill in the missing information to solve for  $s$ .

$$-31 = s - 6$$

$$-31 + \underline{\hspace{2cm}} = s - 6 + \underline{\hspace{2cm}} +$$

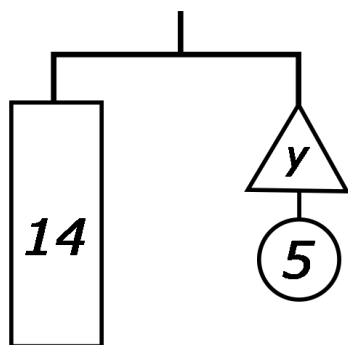
$$\underline{\hspace{2cm}} = s$$

b. Which inverse operation was used to solve the equation?

c. Check your solution.

2. The hanger diagram represents an equation.

a. Write the equation represented by the hanger diagram.



b. What is the value of  $y$ ?



3. Two bar model diagrams are shown.

| Bar Model Diagram 1                                                                                  | Bar Model Diagram 2                                                                                 |
|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <div><div><math>m</math></div><div><div><math>-12</math></div><div><math>-4</math></div></div></div> | <div><div><math>-12</math></div><div><div><math>4</math></div><div><math>m</math></div></div></div> |

a. Write an algebraic equation for each diagram box diagram.

| Algebraic Equation for Diagram 1 | Algebraic Equation for Diagram 2 |
|----------------------------------|----------------------------------|
|                                  |                                  |

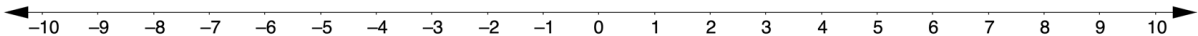
b. Find the value of variable,  $m$ , for each algebraic equation.

| Diagram 1 | Diagram 2 |
|-----------|-----------|
|           |           |

c. Discuss with your partner what you notice about the two equations. Summarize your discussion.

4. Solve the equation using the number line.

a. Draw a representation for  $x - 1 = -6$  on the number line.



b. What is the value of the variable,  $x$ ?

### 12.7.5 Wrap-Up: Reflection

1. Plot a point to indicate how you feel about today's learning target: "I can use diagrams and inverse operations when solving one-step addition and subtraction equations."

